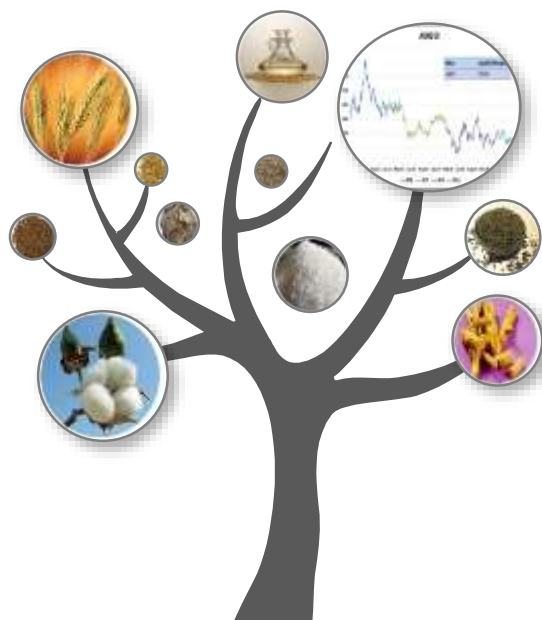




AGRIDEX

INDEX METHODOLOGY

JUNE 2020

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1 Introduction

NCDEX AGRIDEX is India's first return based agricultural futures Index which tracks the performance of the ten liquid commodities traded on NCDEX platform. The index represents the basket of ten commodities that are selected based on the liquidity on the exchange platform. The NCDEX AGRIDEX serves as a benchmark and one can replicate the performance of the underlying commodities. These underlying commodities contracts are mostly deliverable, settled on spot prices, have circuits governing daily price limits and also restrict the size of participation using position limits at the client and member level. Strong surveillance exists both at the exchange and the regulator to prevent any abnormal trading behavior in these commodities. The index, therefore, will be a robust indicator of the broader market.

NCDEX AGRIDEX is computed in real-time using the prices of the 10 most liquid commodities traded on the NCDEX platform. The eligibility of its constituents is decided by the average traded value on the exchange platform and based on other SEBI guidelines. Commodities are weighted based on their national production value and their total traded value on the exchange platform. Due to SEBI guidelines on Futures on Commodity Indices, NCDEX AGRIDEX was rescaled to 1000 on 22nd May 2020.

2 Selection criteria for commodity futures

The Exchange has identified a set of general eligibility criteria for the inclusion of commodities in the index. These criteria are as follows:

- Commodities selection will be based on both endogenous (liquidity criteria) and exogenous (importance in Indian/ Global commodities market) importance. NCDEX has specified that the commodity to be included in the index must be a **physical commodity**. *[Note: Contract on a particular commodity not necessarily required to be settled by physical delivery, for the commodity to be considered a physical commodity].*
- Initially most liquid commodities are proposed for inclusion, however, based on Index Steering Committee may approve the same.
- For a commodity to be eligible for inclusion in the index, futures contracts on the commodity should be in existence for at least the previous 12 months, and should have traded for at least 90% of the trading days during the previous 12 months.

- For a Commodity to be eligible for inclusion in the index, traded volume data for such Commodity must be available for at least 1 year.
- As per SEBI guidelines, commodities having at least 80% of combined weightage in the index must fulfill **ADTV*** ≥ 75 Cr. [Note: ADTV of a commodity will be cumulative of all the symbols at the commodity level.]
- A commodity futures contract that is part of the index would be removed (or replaced), in case of a ban/ suspension/ curb/ regulatory action on commodity trading by the Govt. agency/regulator or other appropriate agencies.
- The removed futures contract would be replaced by the commodity futures that are now eligible for inclusion in Index. The commodities shall be added and removed at the end of the trading session. No changes shall be made in the index during market hours.

3 Weight Calculations

NCDEX AGRIDEX is a value-weighted index. Ten selected commodities are assigned weights based on:

- The total traded value of the commodity on the NCDEX platform in the last 12 months. For a normal rebalancing event, the last 12 months represent Jan-Dec. For untimely rebalancing, the last 12 months are considered. For example, if rebalancing is done on 28 July 2016, then liquidity data from 01 July 2015 to 30 June 2016 is considered.
- The latest available annual value of production data or estimates of production (average of the preceding five years) for the commodity. For example, for calculating weights for FY 2019-20, production data for FY14-15, FY15-16, FY16-17, FY17-18 and FY18-19 are used.

After computing production weights and liquidity weights, both are combined in a 1:1 ratio to calculate the final weights (also called CIPs).

3.1 Production Data Sources

Annual Production Data for the commodities are taken from very reliable sources. The sources are as follows:

- Agricoop: Cereals and Pulses

- Spice Board: Spices
- USDA: Sugar and Soya oil
- State Data: Guar seed

For derived commodities amount of production is estimated as:

$$\text{Guar gum Production} = \text{Guarseed Production} \times 30\%$$

$$\text{Raw Cotton Production} = \text{Cotton Lint} \times 34\%$$

$$\text{Cottonseed Production} = \text{Raw Cotton Production} \times 63.5\%$$

$$\text{Cottonseed oilcake Production} = \text{Cotton Production} \times 80\%$$

3.2 Value of production

The value of production of a commodity is defined as the total monetary value (in INR) of the agricultural commodity produced in the past one fiscal year in India.

The value of production is calculated by multiplying the total quantity produced with the average spot market price of the commodity.

$$\text{Value of Production} = \Sigma (\text{Spot Price of Commodity} \times \text{Production})$$

Price of Commodity is the Average Spot Price from NCDEX spot polling for the Period of Interest.

Assumptions for calculating the Price of Commodity are as follows:

- In case 2 prices are available on the same day, then the latest price available for that day is taken as the spot price for that day.
- In case spot polling is not performed on a particular day, then Spot price for that day is not used in the calculation.
- The average is calculated only for days on which Spot Price is available.

3.3 Commodity Production Percentage

The production value of each commodity (in INR) on the national scale forms the basis of calculation of the commodity Production Percentage. The data for production value is obtained from reliable government sources. Data for the past 5 financial years is obtained for all the 10 commodities in the index and averaged. Final production data or the latest production estimates are used for weight calculations. The CPP for each commodity is calculated as a

percentage of the production value of the particular commodity to the total production value of all the commodities.

The following expression shows the calculation of CPP:

$$CPP(i) = \frac{\text{Value of Production (i)}}{\text{Total Production Value of all commodities in Index}}$$

3.4 Commodity Liquidity Percentage

The notional value of commodity futures traded on the NCDEX platform forms the basis of calculating the commodity liquidity percentage. The sum of the traded value over a period of last 1 calendar year is calculated for each commodity. The Commodity Liquidity Percentage (CLP) for each commodity is given as the percentage of the commodities traded value to the total traded value of all commodities. The following expression shows the calculation of CLP:

$$CLP(i) = \frac{\text{Trade Value (i)}}{\text{Total Trade Value of all commodities in Index}}$$

Where trade value is given by total value (in Crores) traded on the NCDEX platform.

3.5 Commodity Index Percentage

The Commodity Production Percentage and the Commodity Liquidity Percentage are combined in a 1:1 ratio to calculate the commodity Index percentage. A generalized formula for computing the CIP is

$$CIP = (0.5 \times CLP) + (0.5 \times CPP)$$

CLP: Commodity Liquidity Percentage

CPP: Commodity Production Percentage

For example, if commodity ABC has a Production Percentage of 40% and a Liquidity Percentage of 20% its weight in the index will be $(0.5 \times 40) + (0.5 \times 20) = 30\%$.

3.6 Floors and Caps to Commodity Weights

A cap of 20% and a floor of 3% is enforced onto the individual Commodity Index Percentage. The cap ensures that no single commodity dominates the index performance. The floor ensures that each commodity, however small, has a meaningful contribution to the Index. To

ensure diversification, no related group of commodities (sectors) may constitute more than 40% of the weightage in the index. Categories for various commodities are as following:

Table 1: Sector-wise commodity classification

	Sector	Commodities
1	Edible Oils and Oilseeds	RM Seed, Soybean, Refined Soy Oil, Crude palm Oil
2	Pulses	Chana, Moong
3	Industrial Products	Guarseed, Guargum, Castorseed
4	Grains	Wheat, Barley, Maize, Paddy
5	Cotton Complex	Cotton seed oilcake, Kapas, Cotton
6	Spices	Turmeric, Dhaniya, Jeera
7	Softs	Sugar

If a commodity has ADTV < 75 Cr over the last 12 months, its weight in the index is capped at 15% as per SEBI guidelines.

Step 1. Sector-wise Cap

CIPs are calculated from the above-mentioned methodology. Cap/Floor are enforced on these CIPs so that final weights satisfy all the guidelines mentioned in section 2 of the document.

Step 1a: Adjusting sector weight to 40%: If weight for any sector exceeds 40% (over-weighted sectors), the excess is distributed among remaining commodities in proportion of their original/latest (latest means weight at end of the last step) weights, and keeping their individual as well as sectorial cap in check.

Example:

Table 2 Step 1a

Commodity	CIP	Excess = 5.61%	Change	Final CIP
RMSeed	15.42%		1.90%	13.52%
Soybean	21.74%		2.67%	19.07%
Soy Oil	8.45%		1.04%	7.41%
SECTOR	45.61%			40.00%

Step 1b: Individual caps within the over-weighted sectors: If any commodity-specific weight exceeds the cap after this process, the new excess is obtained from the commodity-specific weight, and is distributed among the other commodities in the sector in proportion to their latest weights.

Below example clarifies things further:

Table 3 Step 1b

Commodities	CIP	Excess = 6.50 %	Commodities	Change	Final CIP	Excess = 1.94 %	Commodities	Change	Final CIP
RMSeed	12.50%		RMSeed	1.75%	10.75%		RMSeed	1.15%	11.91%
Soybean	25.50%		Soybean	3.56%	21.94%		Soybean	-1.94%	20.00%
Soy Oil	8.50%		Soy Oil	1.19%	7.31%		Soy Oil	0.79%	8.10%
SECTOR	46.50%		SECTOR		40.00%		SECTOR		40.00%

In the above case, an excess of 6.50% is removed from this sector and will be distributed among other sectors. Excess is removed in proportion (1.75%, 3.56% and 1.19% from RMSeed, Soybean and Soy Oil respectively) to their original weights. In this example, it can be seen that the weight for Soybean breaches the 20% cap, so need 1 additional step is needed in which the excess of 1.94% is distributed to the other 2 commodities proportional to their latest weights.

Step 1c: Individual floors within the over-weighted sectors

The above step is also performed in case a commodity is below the floor. Weights are distributed keeping individual caps and floors in check.

Example:

Table 4 Step 1c

Commodity	CIP	Excess = 2.40 %	Commodity	Change	Final CIP					
RMSeed	17.00%		RMSeed	0.96%	16.04%					
Soybean	23.00%		Soybean	1.30%	21.70%					
Soy Oil	2.40%		Soy Oil	0.14%	2.26%					
SECTOR	42.40%		SECTOR		40.00%					
			Excess = 1.70 %							
			Commodity	Change	Final CIP	Deficit = 0.53 %	Commodity	Change	Final CIP	
			RMSeed	1.49%	17.53%		RMSeed	-0.53%	17.00%	
			Soybean	-1.70%	20.00%		Soybean	0.00%	20.00%	
			Soy Oil	0.21%	2.47%		Soy Oil	0.53%	3.00%	
			SECTOR		40.00%		SECTOR		40.00%	

In the last step, a deficit of 0.53% is taken from RMSeed only as Soybean is at 20% cap level.

For sectors whose weights are below 40%, the step 1 is not applicable.

Example:

Table 5

Commodity	CIP
RMSeed	12.00%
Soybean	23%
Soy Oil	2.40%
SECTOR	37.40%

For this case, step 1 is not applicable and jumps directly to step 2.

Following example shows the implementation of Step 1:

Table 6

BASE DATA		STEP 1		
	CIP		CHANGE	CIP
CASTOR	6.00%	CASTOR		6.00%
COCUD	10.00%	COCUD		10.00%
CORIANDER	2.00%	CORIANDER		2.00%
GUARGUM	6.00%	GUARGUM		6.00%
GUARSEED	21.00%	GUARSEED		21.00%
JEERA	8.00%	JEERA		8.00%
RMSEED	15.00%	RMSEED	-1.36%	13.64%
SOYBEAN	21.00%	SOYBEAN	-1.91%	19.09%
SOY OIL	8.00%	SOY OIL	-0.73%	7.27%
TURMERIC	3.00%	TURMERIC		3.00%
SECTOR	100.00%	SECTOR		96.00%
		EXCESS		4.00%

Step 2 Commodity wise cap and floor

In this step, commodity wise cap and floors are enforced. Any sector for which step 1 was applicable, is not modified in this step.

Step 2a: Individual Caps: Excess left from step 1 is added to any commodity-specific excess and is redistributed among the commodities (which were not a part of step 1 and whose weights are more than the cap itself) keeping the sector-wise cap in check. Applying step 2a to the example given above.

Table 7

STEP 2a			STEP 2a		
	CHANGE	CIP		CHANGE	CIP
CASTOR		6.00%	CASTOR	0.86%	6.86%
COCUD		10.00%	COCUD	1.43%	11.43%
CORIANDER		2.00%	CORIANDER	0.29%	2.29%
GUARGUM		6.00%	GUARGUM	0.86%	6.86%
GUARSEED	-1%	20.00%	GUARSEED	0.00%	20.00%
JEERA		8.00%	JEERA	1.14%	9.14%
RMSEED		13.64%	RMSEED	0.00%	13.64%
SOYBEAN		19.09%	SOYBEAN	0.00%	19.09%
SOY OIL		7.27%	SOY OIL	0.00%	7.27%
TURMERIC		3.00%	TURMERIC	0.43%	3.43%
SECTOR		95.00%	SECTOR		100.00%
EXCESS		5.00%			

Excess from the commodity-specific cap is obtained (1% in this example for Guarseed), taking the total excess to 5%. This 5% is distributed to other commodities in proportional to their original/latest weights.

Step 2b: Individual Floors

In this step, Coriander weight is brought to the floor of 3% and is redistributed to other commodities in proportion to their latest weights.

Table 8

STEP 2b		
	CHANGE	CIP
CASTOR	-0.13%	6.73%
COCUD	-0.22%	11.21%
CORIANDER	0.71%	3.00%
GUARGUM	-0.13%	6.73%
GUARSEED	0.00%	20.00%
JEERA	-0.17%	8.97%
RMSEED	0.00%	13.64%
SOYBEAN	0.00%	19.09%
SOY OIL	0.00%	7.27%
TURMERIC	-0.06%	3.36%
SECTOR		100.00%

The final weights obtained above satisfy all the rules set forth earlier. In case the final weights obtained breach any guideline specified earlier, the steps are repeated.

Final weights obtained are rounded up to 8 decimal places. For example, 0.15232425163 is rounded to 0.15232425.

3.7 Commodity Weights

Table 9 Commodities Weights for FY 2020-21

Commodity	Weightage (%)
SOYBEAN	18.430218
CHANA	17.809556
RMSEED	12.122526
GUARSEED	11.161946
CASTOR	10.089767
COCUD	8.686219
SOY OIL	8.144461
GUARGUM	6.452903
JEERA	3.885969
CORIANDER	3.216436

3.8 Commodities contract calendar

This is the contract calendar for different commodities which are part of NCDEX AGRIDEX for next 6 months.

Table 10 Contract calendar for commodities

Commodities/Month	Jun	Jul	Aug	Sep	Oct	Nov	Dec
CASTOR	Jul	Aug	Sep	Oct	Nov	Dec	Jan
CHANA	Jul	Aug	Sep	Oct	Nov	Dec	Jan
COCUD	Jul	Aug	Dec	Dec	Dec	Dec	Apr
DHANIYA	Jul	Aug	Sep	Oct	Nov	Dec	Jan
GUARGUM	Jul	Aug	Sep	Oct	Nov	Dec	Jan
GUARSEED	Jul	Aug	Sep	Oct	Nov	Dec	Jan
JEERA	Jul	Aug	Sep	Oct	Nov	Dec	Jan
RM SEED	Jul	Aug	Sep	Oct	Nov	Dec	Jan
SOYBEAN	Jul	Aug	Sep	Oct	Nov	Dec	Jan
SOY_OIL	Jul	Aug	Sep	Oct	Nov	Dec	Jan

4 Index Construction Methodology

NCDEX AGRIDEX tracks the value of the hypothetical investment portfolio starting at 1000 on the day of the launch of the index. This hypothetical investment portfolio is reset on launch day using the weights as per CIPs, which are explained in the previous section. We define separate commodity-specific indices as the sum of value invested in front contract and value invested in the next contract.

$$Comm_Index_t^c = V_t^f + V_t^n$$

At the time of launch of the index,

$$V_t^f = 1000 \text{ and } V_t^n = 0.$$

NCDEX AGRIDEX is calculated from there Commodity Indices as the weighted sum of these commodity indices weighted by their CIPs:

$$Index_t = \sum_c Comm_Index_t^c * CIP_c$$

At $t = 0$, Commodity Indices for all commodities start from 1000.

$$Comm_Index_0^c = V_t^f + V_t^n = 1000 + 0 = 1000$$

$$Index_0 = \sum_c 1000 * CIP_c = 1000$$

During the non-roll period (after the expiry of contract), $V_t^n = 0$, and V_t^f is calculated as:

$$V_t^f = V_{t-1}^f * \frac{P_t^f}{P_{t-1}^f}$$

Here, P_t^f denotes the front month price at time t . P_{t-1}^f is the daily settlement price for front contract on the previous day. P_t^f refers to the last traded price for the front contract on the current day for calculation of the index on the current day. For calculation of the closing price of the index, P_t^f refers to the daily settlement price for the front contract on the current day.

4.1 Roll-over Methodology

During roll-over, the hypothetical portfolio that the index represents is shifted from expiring front contracts (which are expiring in the current month) to next available contracts over a period of 3 days. On the first day of a roll-over, one-third of investment in the front contract is shifted to the next contract. On the second day of a roll-over, half of the remaining investment in the expiring front contract is shifted to the next contract. On the last day of a roll-over, all of the remaining investment in the expiring front contract is shifted to the next contract. Roll-over is done on the first 3 business days of the month.

Value invested in time 't' in front and next contracts respectively are calculated using:

$$V_t^f = \left(V_{t-1}^f * (1 - RW_t) \right) * \frac{P_t^f}{P_{t-1}^f}$$

$$V_t^n = \left(V_{t-1}^f * RW_t + V_{t-1}^n \right) * \frac{P_t^n}{P_{t-1}^n}$$

Where,

RW : Roll Weight.

$$\begin{aligned} RW &= 1/3 && \text{for the first day of roll - over} \\ &= 1/2 && \text{for the second day of roll - over} \\ &= 1 && \text{for the last day of roll - over} \end{aligned}$$

After rollover and before the expiry of the contract, $V_t^f = 0$, and V_t^n is calculated as:

$$V_t^n = V_{t-1}^n * \frac{P_t^n}{P_{t-1}^n}$$

An example of calculation for commodity index is provided below

Example (commodity-specific):

Table 11 Index Calculation during Roll-Over Period

Day	RW	Prices (Contract)		V_front	V_next	Index
		Front	Next			
0		750	800	1000	0	1000
1		760	810	$V_1^f = V_0^f * \frac{P_1^f}{P_0^f} = 1000 * \frac{760}{750} = 1013.33$	0	1013.33
2	0.33	740	785	$V_2^f = V_1^f * (1 - RW) * \frac{P_2^f}{P_1^f}$ $= 1013.33 * (1 - 0.33) * \frac{740}{760} = 657.78$	$V_2^n = (V_1^f * RW + V_1^n) * \frac{P_2^n}{P_1^n}$ $= (1013.33 * 0.33 + 0) * \frac{785}{810} = 327.35$	985.13
3	0.5	765	805	$V_3^f = V_2^f * (1 - RW) * \frac{P_3^f}{P_2^f}$ $= 657.78 * (1 - 0.5) * \frac{765}{740} = 340$	$V_3^n = (V_2^f * RW + V_2^n) * \frac{P_3^n}{P_2^n}$ $= (657.78 * 0.5 + 327.35) * \frac{805}{785} = 672.96$	1012.96
4	1	790	825	0	$V_4^n = (V_3^f * RW + V_3^n) * \frac{P_4^n}{P_3^n}$ $= (340 * 1 + 672.96) * \frac{825}{805} = 1038.13$	1038.13
5		775	815	0	$V_5^n = (V_4^n) * \frac{P_5^n}{P_4^n} = 1038.13 * \frac{815}{825} = 1025.54$	1025.54
T		P_T^f	P_T^n	0	$V_T^n = (V_{T-1}^n) * \frac{P_T^n}{P_{T-1}^n}$	Index Value _T
T+1		P_{T+1}^f	P_{T+1}^n	$V_{T+1}^f = (V_T^n) * \frac{P_{T+1}^f}{P_T^n}$	0	Index Value _{T+1}

*Where T denotes expiry date of the front contract

Post Expiry next month becomes front-month contract, V^n becomes V^f . Similarly, Prices P_t^n will be now calculated as P_t^f . Where,

P_t^n = Price of next month contract at the expiry date

P_t^f = Price of front-month contract at the expiry date

Example (Multiple commodities):

	RM Seed		Soybean		Soy oil	
CIP	0.25		0.35		0.4	
Date	Near	Mid	Near	Mid	Near	Mid
4/29/2020	4215	4141	3786	3810	824	787
4/30/2020	4230	4174	3992	3826	825	797
5/4/2020	4135	4148	3798	3734	783	751
5/5/2020	4089	4098	3806	3744	772	745
5/6/2020	4123	4133	3824	3756	763	736
5/7/2020	4154	4165	3844	3768	766	737
5/8/2020	4431	4484	4043	4120	656	666

**Prices taken in the examples are not real

$$Index = \sum CIP_i * (V_{fi} + V_{ni}) = 0.25(V_{RMSeed}) + 0.40(V_{Soybean}) + 0.35(V_{Soyoil})$$

	RM Seed		Soybean		Soy_oil		Index
CIP	25%		40%		35%		
Date	Vf	Vn	Vf	Vn	Vf	Vn	
4/29/2020	1000.00	0.00	1000.00	0.00	1000.00	0.00	1000.00
4/30/2020	1003.56	0.00	1054.41	0.00	1001.21	0.00	1023.08
5/4/2020	981.02	0.00	1003.17	0.00	950.24	0.00	979.11
5/5/2020	646.74	323.06	670.19	335.29	624.60	314.22	973.22
5/6/2020	326.06	651.95	336.68	672.53	308.66	618.95	972.85
5/7/2020	0.00	985.58	0.00	1012.43	0.00	928.86	976.47
5/8/2020	0.00	1061.07	0.00	1107.01	0.00	839.38	1001.86

$$Index = 0.25 * (646.74 + 323.06) + 0.40 * (670.19 + 335.29) + 0.35 * (624.60 + 314.22)$$

4.2 Rebalancing

The composition of the index is rebalanced after the financial year has ended i.e. on the first trading day of the new financial year. The commodities that qualify the selection criteria mentioned in section 2 of the document will be included in the index. The weights of the selected commodities will be recalculated. Once the changes in weights are approved by the Index Steering Committee, the new composition of the Index is publicly announced (by the first trading day of January).

The formula for the index calculations changes due to rebalancing.

Let the Index value before the start of rebalancing is $Index_0$.

Normalization factor,

$$NF = \frac{Index_0}{1000}$$

Commodity indices for all commodities are reset back to 1000. The new Index formula becomes:

$$Index_t = NF * \sum_c Comm_Index_t^c * CIP_c$$

This Normalization Factor remains constant throughout the year.

4.3 Conversion Ratio

Conversion Ratio shows that how many number of lots individual commodity contains in one lot of Index.

$$Conversion_Ratio_c^f = NF * \left\{ \frac{V_t^f * CIP_c * 500}{P_t^f * \left(\frac{LotSize_c}{BaseValue_c} \right)} \right\}$$

$$Conversion_Ratio_c^n = NF * \left\{ \frac{V_t^n * CIP_c * 500}{P_t^n * \left(\frac{LotSize_c}{BaseValue_c} \right)} \right\}$$

Here, $LotSize$ is the quantity per lot. For example, 1 lot of Ref Soy Oil is for 10 MT. $BaseValue$ is a quotation unit. Ref Soy Oil is quoted per 10 kg.

One lot of Index has 500 units. Therefore, at the price of 1000, the index contract value is Rs. 5,00,000.

Example:

Conversion Ratio for Day1 (see Table 9 section 4.1) where the front-month contract price is 760 and next month's contract price is 860. Suppose CIP = 0.25, and normalization factor = 1.

$$Conversion_Ratio_c^f = 1 * \left\{ \frac{1013.33 * .25 * 500}{760 * \left(\frac{10 MT}{10 Kg} \right)} \right\}$$

$$Conversion_Ratio_c^f = 0.1667$$

$$Conversion_Ratio_c^n = 1 * \left\{ \frac{0 * 0.25 * 500}{810 * \left(\frac{10 MT}{10 Kg} \right)} \right\}$$

$$Conversion_Ratio_c^n = 0$$

4.4 Market Disruption events

Events that are not expected in the normal course of business will qualify as market disruption events. An example of such an event could be a ban/suspension of futures trading on the commodity by a regulatory organization. The Index Steering Committee, responsible for index maintenance, will be the deciding authority in all such events and will take decisions related to index reconstitution and rebalancing.

In the event of such activity happens the following immediate actions shall be taken:

- Reselection of commodities (excluding banned/suspended commodities) (the logic for reselection shall be the same).
- Recalculation of weights (using the same logic)
- Recalculation of the normalization factor

Index changes shall be implemented only after market hours. No change in the process will be made during market hours.

For any queries, please contact:

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